



YESHWANTRAO CHAVAN COLLEGE OF ENGINEERING



Department of Computer Technology

PQAI-2 Seminar on

Advanced monitoring and control mechanism for electric vehicle charging station using Machine learning

PRESENTED BY: Group CT- 23

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COLLEGE GUIDE:

Prof.N.U.Sambhe

SECTION : A & B

YEAR: 2025-26

INDUSTRIAL CO-GUIDE:

Prof.P.S.Shete, EL Dept

Department Vision and Mission

Vision

To be a well-known center for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary arena by developing problem-solving skills through emerging technologies.

Program Outcomes/Program Specific Outcomes

At the end of Program, Graduate students will have

- PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
- PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
- PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
- PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
- PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

Program Outcomes/Program Specific Outcomes

- PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).
- PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9).
- PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
- PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences
- PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

Program Outcomes/Program Specific Outcomes

- PO11: Lifelong Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies
- [PSO 1] An ability to acquire skills to design & develop quality software using cutting-edge technologies of Computer Engineering.
- [PSO 2] An ability to use knowledge of various domains as per thrust areas to formulate and implement ideas for providing innovative solutions.

Course Outcomes

Course Outcomes	Program Outcomes											Program Specific Outcomes	
	1	2	3	4	5	6	7	8	9	10	11	PSO 1	PSO 2
Identify a problem, analyze research gaps through a literature survey, and propose an ethical, innovative solution with societal impact.	3	3	3	3	3	2	2	3	3		2	3	2

Course Outcomes	Program Outcomes											Program Specific Outcomes	
	1	2	3	4	5	6	7	8	9	10	11	PSO 1	PSO 2
Develop a structured project report detailing the problem, methodology, proposed solution, and findings, adhering to professional and ethical standards.	3	3	3	3	3	3	2	3	3		2	3	3
Implement a basic prototype/model to validate the proposed solution, demonstrating practical application of knowledge and technical skills.	3	3			3			3	3	3	3	3	3
Avg	3	3	3	3	3	2.5	2	3	3	3	2.3	3	2.6667

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Problem Statement

A Machine Learning system for EV charging stations that predicts optimal current, detects faults in real time, and optimizes operations for efficient and seamless charging.

Introduction

As electric vehicles (EVs) become an integral part of our daily lives, the need for smarter and more efficient charging infrastructure has never been greater. Charging stations must do more than just provide power — they need to be reliable, adaptable, and capable of meeting growing demand while minimizing energy waste.

Our project tackles this challenge by creating a predictive model using a data-driven approach and supervised machine learning. By analyzing both historical and real-time data, the model can forecast charging demand at public stations, allowing energy to be distributed efficiently, reducing grid stress, and enhancing the user experience.

But it doesn't stop there. The project also integrates advanced monitoring and control mechanisms, enabling real-time fault detection and better energy management. These features ensure charging stations remain efficient, safe, and capable of adapting to fluctuating energy loads and user needs.

Objectives

Primary Objective/Problem Definition:

To design and develop a data-driven predictive model using supervised machine learning regression methods to accurately forecast charging demand at public charging stations, and to propose advanced monitoring and control mechanisms to optimize energy management, reduce peak demand, and enhance the overall charging experience.

Specific Objectives:

1.To Collect Dataset: Gather historical data on electric vehicle charging patterns, energy consumption, and other relevant factors to support predictive modeling.

2.To design and develop model for Predictive analysis:Develop a machine learning-based predictive model that utilizes collected data to forecast charging demand and energy consumption.

3.To monitor and Predict: Design a real-time monitoring system that utilizes predictive analytics to forecast charging demand, detect anomalies, and optimize charging station operations.

4.To detect and manage fault: Develop a fault detection and management system that uses automated control mechanism to identify potential faults, alert operators, and provide recommendations for maintenance and repair.

Work Done

1. Improved and Interactive User Interface (UI)

- Designed a user-friendly and engaging front-end.
- Integrated dashboard to display continuous outputs and variations.

2. Continuous Automated Data Processing & Visualization

- Shifted from manual input to automated continuous input for real-time monitoring.
- Dashboard maps outputs continuously to analyze performance trends.
- Visualization of variation trends helps in identifying critical patterns.

3. Fault Detection Mechanism

- Implemented a threshold-based fault detection system.
- Voltage regulation errors are flagged when values exceed or fall below thresholds.
- Future scope: Enhance fault detection with advanced ML techniques and integrate per-user/vehicle monitoring for better data analysis and security.

4. **Research Paper & Patent Work**

- Prepared a full research paper from scratch on ML-based EV monitoring.
- Paper approved by guide and currently under publication process.
- Patent form already filed for innovative aspects of the project.

Expected Input and Output

EV Charging Station - Dashboard

FAULT

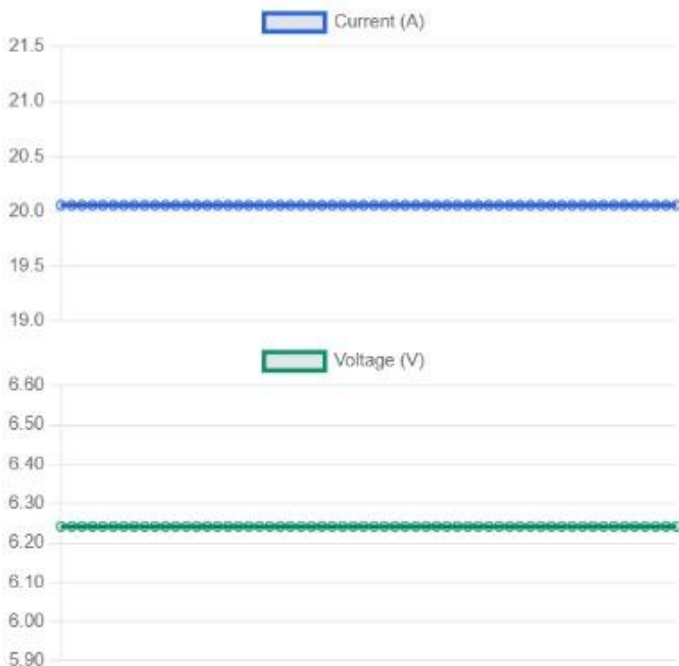
I=20.056A, V=6.242V, T=415.601C

Persistent fault detected. Monitoring stopped.

Thresholds

```
{
  "id": 1,
  "min_current": 0,
  "max_current": 60,
  "min_voltage": 180,
  "max_voltage": 260,
  "max_temperature": 80
}
```

Charts



Recent Events

- 2025-09-17T14:19:22.580088+00:00 - INFO - ALERT: Persistent fault detected
- 2025-09-17T14:19:22.579089+00:00 - FAULT_DETECTED - Voltage out of range: 6.242; Temperature too high: 415.601
- 2025-09-17T14:19:22.579089+00:00 - CHARGING_STOPPED - Voltage out of range: 6.242; Temperature too high: 415.601
- 2025-09-17T14:13:37.649174+00:00 - INFO - ALERT: Persistent fault detected
- 2025-09-17T14:13:37.648135+00:00 - CHARGING_STOPPED - Voltage out of range: 6.242; Temperature too high: 415.601
- 2025-09-17T14:13:37.645935+00:00 - FAULT_DETECTED - Voltage out of range: 6.242; Temperature too high: 415.601

Status of Research Paper /Patent

FORM 2 THE PATENT ACT 1970 (39 of 1970) & The Patents Rules, 2003 PROVISIONAL/COMPLETE SPECIFICATION (See section 10 and rule13)	
1. TITLE OF THE INVENTION: Advanced Monitoring and Control Mechanism for Electric Vehicle Charging Station using Machine Learning	
2. APPLICANT (S) (a) NAME: <u>Prof. N. U. Sambhe</u> (b) NATIONALITY: <u>Indian</u> (c) ADDRESS: <u>YCCE Nagpur</u>	
3. PREAMBLE TO THE DESCRIPTION	
PROVISIONAL EV charger uses machine learning to optimize power distribution and monitor faults in EV charging stations. It ensures safe operation by dynamically managing load and controlling charger connectivity.	COMPLETE This invention relates to the field of electric vehicle (EV) infrastructure, more specifically to a smart system that optimizes the monitoring and control of EV charging stations using machine learning algorithms.
4. DESCRIPTION <u>EVOptima</u> is an intelligent EV charging station system that uses machine learning to monitor and optimize power distribution across multiple charging points. It predicts electrical parameters, detects faults, and dynamically manages transformer load to ensure safety and efficiency.	
5. CLAIMS A machine learning-based system for optimizing power distribution in EV charging stations, adjusting current in real-time to prevent overloads. The system detects faults and automatically disconnects faulty chargers to maintain efficient operation.	
6. DATE AND SIGNATURE	

Advanced Monitoring and Control Mechanism for Electric Vehicle Charging Station Using Machine Learning

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Abstract

With the rising use of electric vehicles (EVs) globally, the demand for smart, scalable, and efficient charging infrastructure is increasing. Present charging systems are short of handling variable loads, fault detection, and optimum energy distribution. To meet these demands, this paper presents an innovative framework that includes real-time monitoring, intelligent fault detection, prediction analytics, and adaptive control by machine learning (ML) algorithms. Our approach utilizes supervised ML models to forecast energy demand, optimize charging times, and generally enhance security. It is built on a cloud platform with IoT sensors and auto-decision mechanisms to deliver an integral and sustainable charging experience. The solution also considers smart cities and is centred on user-friendliness, operational reliability, and sustainability. Through extensive testing, we demonstrate significant advancements in energy efficiency, system robustness, and service quality at EV charging stations.

Index Terms— Electric Vehicles, Smart Charging, Machine Learning, Energy Management, Fault Detection, Predictive Analytics, Internet of Things (IoT), Smart Grids, Cloud Integration.

Introduction

Electric vehicles (EVs) have transformed the transport industry on the premise of limiting greenhouse gas emissions as well as lessening the consumption of fossil fuels. Also, while consumer rates increase at a high rate, there is a necessity for efficiency and scalability in EV charging stations. However, the majority of traditional charging stations are beset with issues of low utilization rate, wastage of energy, lower quality of service and surprise system failure. All these problems call for more provision of good monitoring and control systems. Besides increasing EV charging needs that come with boundless challenges, machine learning (ML) is among numerous solutions within our reach. It can predict trends independently and assist in sound decision-making

processes to deal with and control real-time charging information and set energy rates. The computerized processing of large amounts of data generated through charge incidents, can create patterns that can guide resources according to projected demand, without restricting usage by no-shows or rogue patronage. This article presents a future-proof model of charging employing machine learning, systems monitoring in real-time, and fault detection through an engagement model based on the cloud to establish a completely dynamic and independent ecosystem. With this system, we hope to offer an improved, safer, and more convenient charging experience for the end-user, which will be part of our broader visions of green cities and clean transportation.

References(*Strictly in IEEE format)

1. Almaghrebi, A.; Aljuheshi, F.; Rafaie, M.; James, K.; Alahmad, M. Data-Driven Charging Demand Prediction at Public Charging Stations Using Supervised Machine Learning Regression Methods. *Energies* **2020**, *13*, 4231. <https://doi.org/10.3390/en13164231>
2. Shufian, Abu & Hannan, Nasif & Emon, Mukter Hossain & Kabir, Shaharier & Fattah, Shaikh Anowarul & Hossain, Md Sajid. (2024). Automatic Fault Detection and Analysis Of Electric Vehicle Charging Station by Machine Learning. 10.1109/TENSYMP61132.2024.10752256.
3. S. Shahriar, A. R. Al-Ali, A. H. Osman, S. Dhou and M. Nijim, "Prediction of EV Charging Behavior Using Machine Learning," in *IEEE Access*, vol. 9, pp. 111576-111586
4. K. N. B. S. Hitesh, P. S. Teja and J. R. Kumar, "Electric Vehicle Charging Demand Prediction using Machine Learning Algorithms," *2024 8th International Conference on Inventive Systems and Control (ICISC)*, Coimbatore, India, 2024, pp. 47-54, doi: 10.1109/ICISC62624.2024.00015.
5. Aghsaee, R.; Hecht, C.; Schwinger, F.; Figgner, J.; Jarke, M.; Sauer, D.U. Data-Driven, Short-Term Prediction of Charging Station Occupation. *Electricity* **2023**, *4*, 134-153. <https://doi.org/10.3390/electricity4020009>
6. S. M. S and R. Gunabalan, "Enhancement of Charging Efficiency of Batteries for Electric Vehicles: Review," 2022 Fourth International Conference on Emerging Research in Electronics, Computer Science and Technology (ICERECT), Mandya, India, 2022, pp. 1-6, doi: 10.1109/ICERECT56837.2022.10060243.
7. S. Deb, "Machine Learning for Solving Charging Infrastructure Planning: A Comprehensive Review," *2021 5th International Conference on Smart Grid and Smart Cities (ICSGSC)*, Tokyo, Japan, 2021, pp. 16-22, doi: 10.1109/ICSGSC52434.2021.9490407.
8. M. Straka, R. Carvalho, G. V. D. Poel and L. Buzna, "Analysis of Energy Consumption at Slow Charging Infrastructure for Electric Vehicles," in *IEEE Access*, vol. 9, pp. 53885-53901, 2021, doi: 10.1109/ACCESS.2021.3071180.
9. A. Shufian, N. Hannan, M. M. Hossain Emon, S. Kabir, S. A. Fattah and M. S. Hossain, "Automatic Fault Detection and Analysis Of Electric Vehicle Charging Station by Machine Learning," *2024 IEEE Region 10 Symposium (TENSYMP)*, New Delhi, India, 2024, pp. 1-6, doi: 10.1109/TENSYMP61132.2024.10752256.
10. R. Atallah *et al.*, "Quality of Service Evaluation and Forecast for EV Charging Based on Real-World Data," *2023 19th International Conference on Wireless and Mobile Computing, Networking and Communications (WiMob)*, Montreal, QC, Canada, 2023, pp. 280-285, doi: 10.1109/WiMob58348.2023.10187855.
11. S. Shahriar, A. R. Al-Ali, A. H. Osman, S. Dhou and M. Nijim, "Machine Learning Approaches for EV Charging Behavior: A Review," in *IEEE Access*, vol. 8, pp. 168980-168993, 2020, doi: 10.1109/ACCESS.2020.3023388.
12. M. Shukla, D. K. Singh, A. Yadav, A. K. Singh, N. A. Jumaa and A. Mohammed, "Optimizing Electric Vehicle Charging Station Placement in Urban Areas: A Data-Driven Approach," *2023 3rd International Conference on Advance Computing and Innovative Technologies in Engineering (ICACITE)*, Greater Noida, India, 2023, pp. 2835-2838, doi: 10.1109/ICACITE57410.2023.10183075.
13. A. Palaniappan, P. Bhukya, S. K. Chitti and J. Gao, "Data-Driven Analysis of EV Energy Prediction and Planning of EV Charging Infrastructure," *2023 IEEE Ninth International Conference on Big Data Computing Service and Applications (BigDataService)*, Athens, Greece, 2023, pp. 17-24, doi: 10.1109/BigDataService58306.2023.00009
14. M. Straka, R. Carvalho, G. V. D. Poel and L. Buzna, "Analysis of Energy Consumption at Slow Charging Infrastructure for Electric Vehicles," in *IEEE Access*, vol. 9, pp. 53885-53901, 2021, doi: 10.1109/ACCESS.2021.3071180

Thank You